

Experiment No.	04
Experiment Title	Binary Classification using Linear and Kernel-Based Models
GitHub Repository:	https://github.com/The-F00L-00/ML_Lab
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1 Objective

To classify emails as spam or ham using Logistic Regression and Support Vector Machine (SVM) classifiers and to analyze the effect of hyperparameter tuning on classification performance.

2 Problem Statement

For the **Spambase** dataset that contains numerical features extracted from email content and a binary label indicating **Spam or non-spam (ham)**. Implement Logistic Regression and reduce overfitting by **Regularization**. Tune with Grid Search CV and RandomizedSearchCV.

3 Dataset Description

Dataset Name	Spambase
Dataset Source	https://www.kaggle.com/datasets/somesh24/spambase
Number of Samples	4601
Number of Features	58
Number of Classes	2
Missing Values	0
Train-Test Split	Train: 3680 and Test: 921

4 Exploratory Data Analysis

Class Distribution

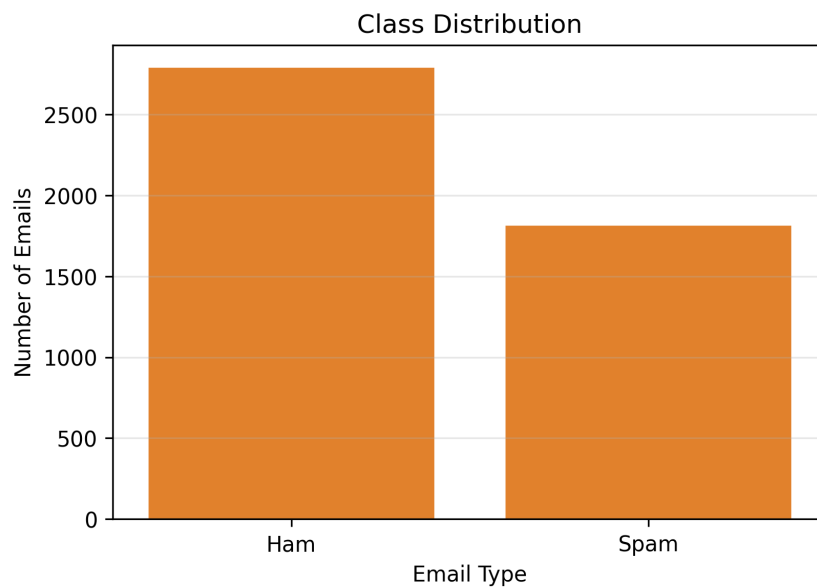


Figure 1: Class Distribution of Emails

Inference:

The class distribution plot shows that the dataset contains more ham emails than spam emails. Approximately 60.6% of the samples belong to the ham class, while 39.4% belong to the spam class. Although the dataset is not perfectly balanced, the class imbalance is moderate and suitable for training classification models without extensive resampling.

Correlation Matrix

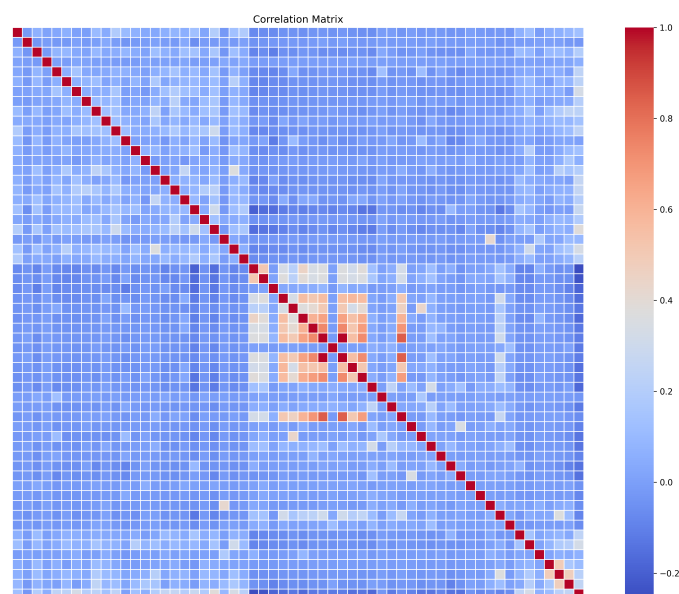


Figure 2: Correlation Matrix of Features

Inference:

The correlation matrix illustrates the relationships among the numerical features in the Spambase dataset. Most features exhibit weak to moderate correlations, while a few groups of features show stronger positive correlations. This indicates that the dataset contains diverse information with limited redundancy, making it suitable for spam classification.

5 Data Preprocessing

- **Missing Value Handling:**

The Spambase dataset contains no missing values; therefore, no imputation or removal of records was required.

- **Encoding:**

The target variable is already represented as binary labels (Spam/Ham), so no additional encoding was necessary.

- **Feature Scaling:**

All numerical features were standardized using StandardScaler. Feature scaling improves the convergence of Logistic Regression and is essential for Support Vector Machine since it is distance-based.

- **Train-Test Split:**

The dataset was divided into training and testing sets using an 80:20 ratio, resulting in 3680 training samples and 921 testing samples.

- **Feature Engineering:**

No additional feature engineering techniques were applied since the dataset already contains extracted numerical features.

6 Machine Learning Algorithm

6.1 Working Principle

- **Logistic Regression:**
Logistic Regression is a probabilistic classification algorithm used for binary classification problems. It models the probability that a sample belongs to a particular class using the **sigmoid function**.
- **Support Vector Machine (SVM)** Support Vector Machine is a margin-based classifier that finds an optimal hyperplane separating two classes by maximizing the margin between them. **Key Idea:** Only a subset of training points (Support vectors) define the decision boundary.

Mathematical Formulation

$$P(y = 1|x) = \frac{1}{1 + e^{-(w^T x + b)}}$$

A threshold (usually 0.5) is applied to convert probability into class labels.

Loss Function: Logistic Regression minimizes the **log-loss (cross-entropy loss)**, which penalizes incorrect predictions.

6.2 Advantages

- Fast and Simple.
- Works well with linearly separable data.
- **Less prone to overfitting:** with L1/L2 Regularization.
- Produces probability estimates for binary classification.

6.3 Limitations

- Assumes Linear Decision boundary.
- Sensitive to outliers.
- Sensitive to multicollinearity.
- Poor performance if classes are not linearly separable.

7 Experimental Setup

Python Version	3.13.13
Scikit-Learn Version	1.7.5
IDE	Jupyter Notebook
Operating System	Fedora 42
Hardware	Intel Core i5 (11th gen), 8GB RAM

8 Hyperparameter Selection

8.1 GridSearchCV for LogisticRegression

Parameter	Value
C	10
solver	liblinear

GridSearchCV for SVM

Parameter	Value
C	1
gamma	scale
kernel	RBF
degree	2

9 Results

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
LogisticRegression	0.9196	0.9316	0.8743	0.9021	0.97
Support Vector Machine	0.925	0.934	0.8846	0.9090	1.00

Table 2: Comparing performance of LogisticRegression and SVM

Kernel	accuracy	F1
Linear	0.925	0.9090
Poly	0.764	0.629
RBF	0.934	0.9206
Sigmoid	0.889	0.8664

Table 3: Comparing the performance of various kernels of SVC

Model	Train accuracy	Test accuracy
Logistic regression	0.9277	0.9196
SVM	0.945	0.9348

Table 4: Comparing Test and train accuracies of both classification models.

10 Result Visualization

10.1 Logistic Regression

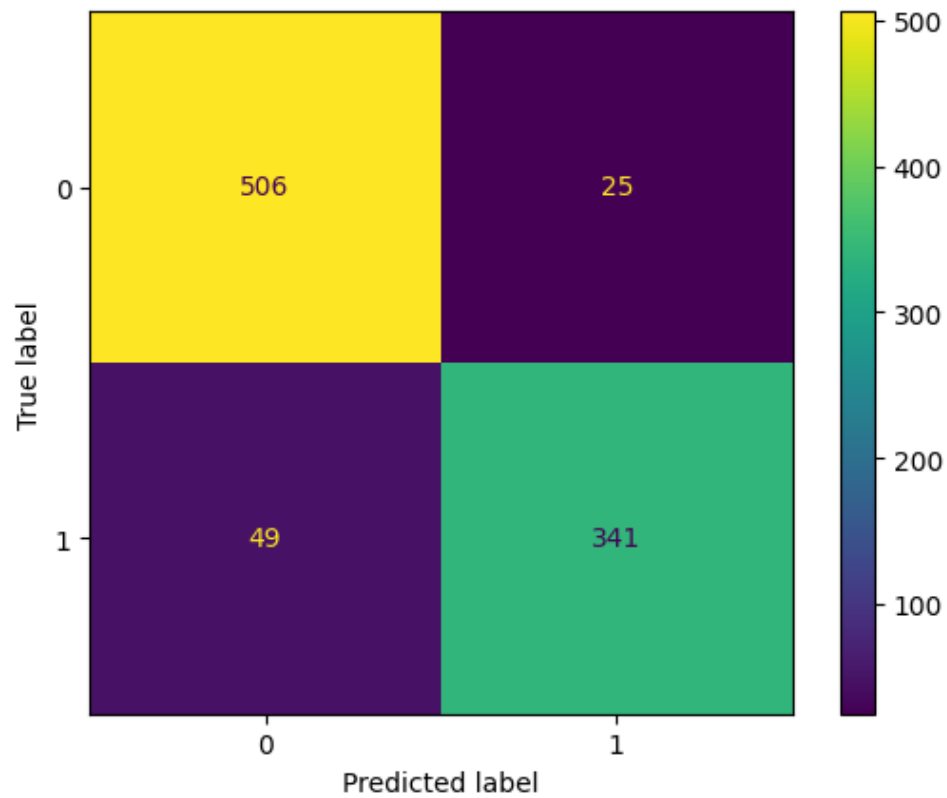


Figure 3: Confusion Matrix

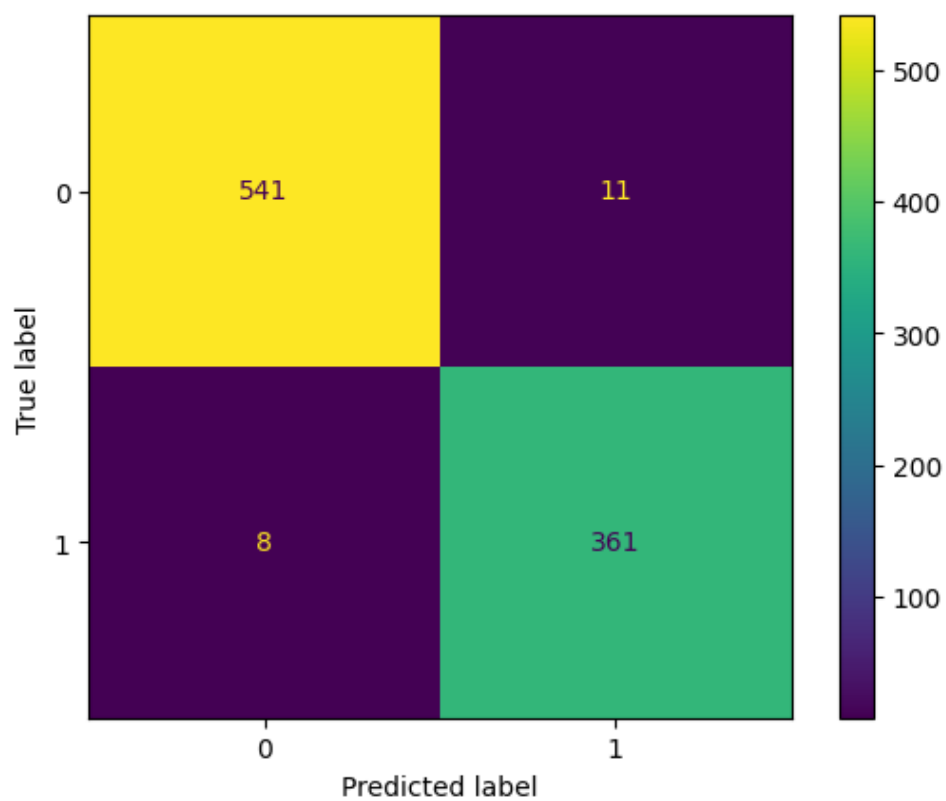


Figure 4: Confusion Matrix Tuned with GridSearchCV.

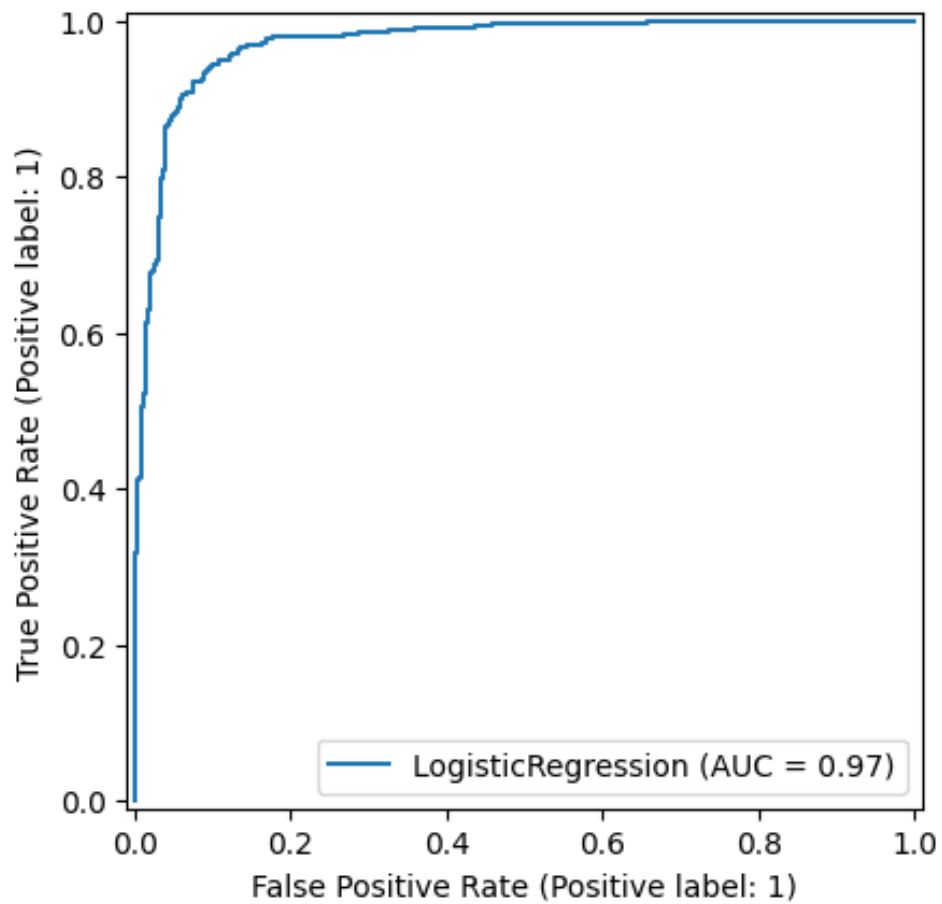


Figure 5: ROC curve

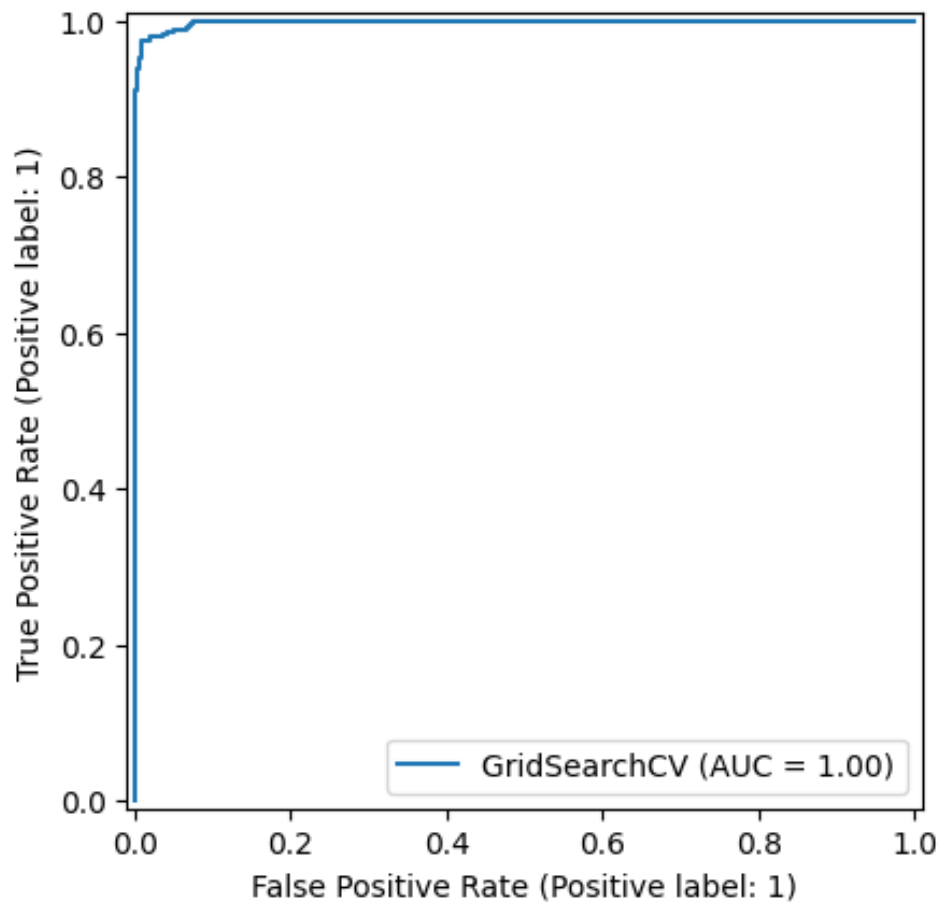


Figure 6: ROC curve Tuned with GridSearchCV.

10.2 SVM

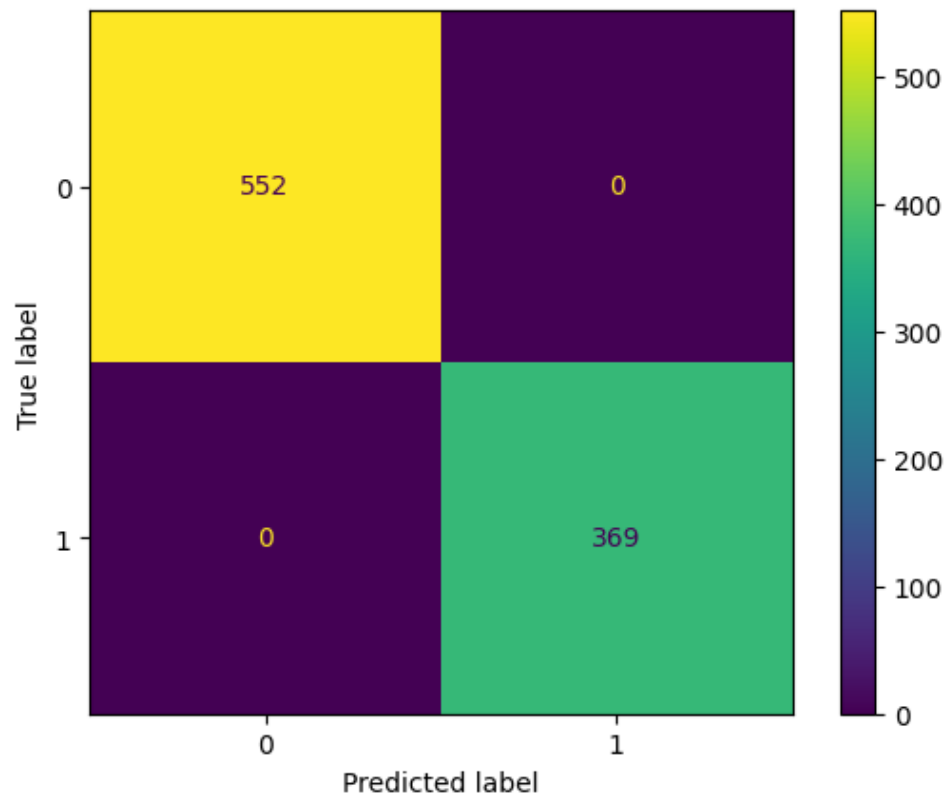


Figure 7: Confusion Matrix

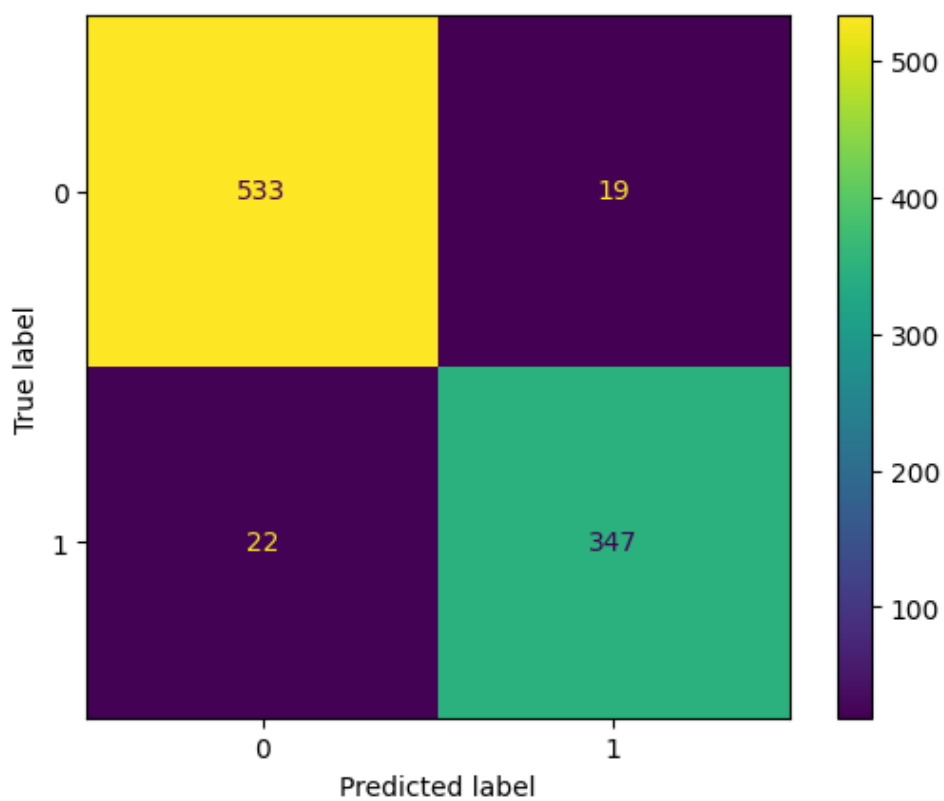


Figure 8: Confusion Matrix Tuned with GridSearchCV.

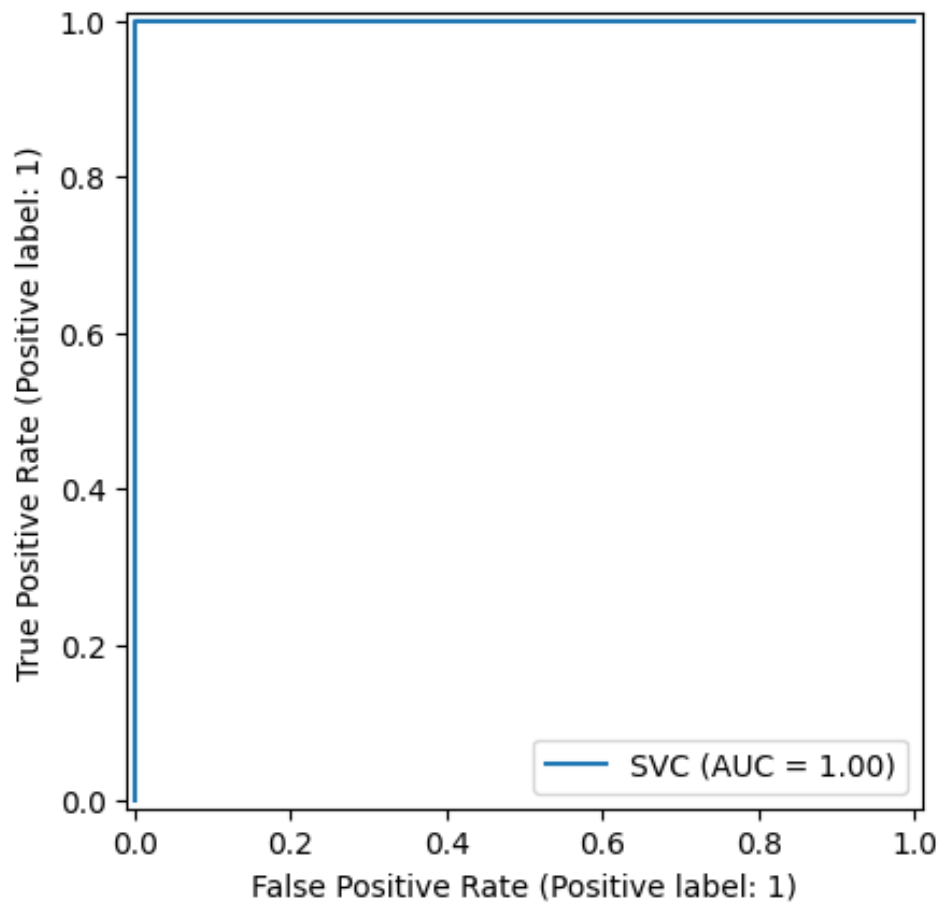


Figure 9: ROC curve

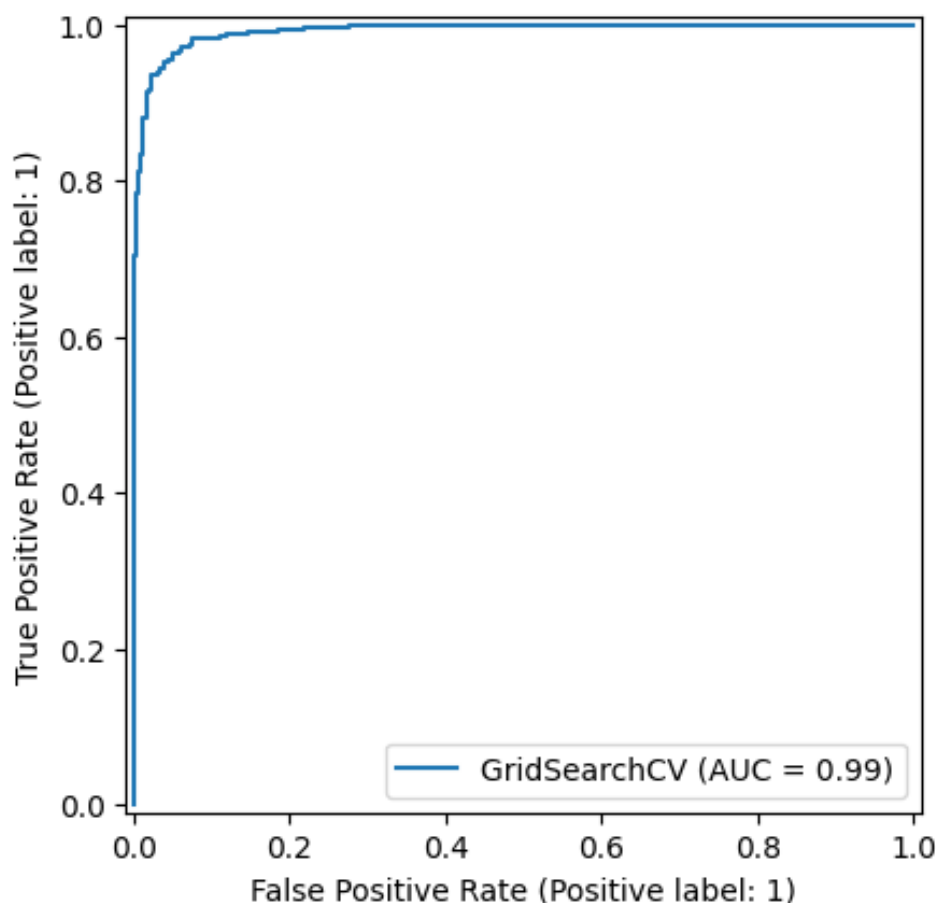


Figure 10: ROC curve tuned with gridSearchCV.

11 Discussion

Since I have trained, GridSearchCV for LogRegression with `max_iter 10000` it takes nearly 1min 10secs to finish. Also GridSearchCV for SVM with `max_iter 5000` that takes 2mins roughly to finish. So these time complexities are the limitations of my implementation.

12 Conclusion

From this experiment, I have learnt to implement Logistic Regression and Support Vector Classification for a given dataset having binary class target. Also tuned them using GridSearchCV, plotted Confusion matrix and ROC curve for all 4 models. Compared Performance metrics for all 4 kernels of SVM.

13 References

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3. T. Hastie, R. Tibshirani and J. Friedman, *The Elements of Statistical Learning*, 2nd ed., Springer, 2009.
4. C. M. Bishop, *Pattern Recognition and Machine Learning*, Springer, 2006.
5. A. Géron, *Hands-On Machine Learning with Scikit-Learn, Keras and TensorFlow*, 3rd ed., O'Reilly, 2022.